

The answer key for the **Individual** round is below.

p	Answer	p	Answer
1	420	6	$\frac{37}{4}$
2	$\frac{1}{16}$	7	8
3	43	8	546
4	π	9	$\frac{168}{5}$
5	15	10	1107

Problem 1. Let $M = 2, N = 3, Y = 5$, and $O = 7$. Compute $M \cdot N \cdot Y \cdot M \cdot O$.

Proposed by: Angie Huang

Answer: $\boxed{420}$

Solution: We have $2 \cdot 3 \cdot 5 \cdot 2 \cdot 7 = \boxed{420}$.

Problem 2. Anna flips a fair coin 4 times. What is the probability it comes up heads every time? Express your answer as a common fraction in simplest form.

Proposed by: Kaining Zhang

Answer: $\boxed{\frac{1}{16}}$

Solution: The probability of a single coin coming up heads is $\frac{1}{2}$. Since each coin flip is independent, our answer is $(\frac{1}{2})^4 = \boxed{\frac{1}{16}}$.

Problem 3. The side lengths of right triangle ABC are all positive integers less than 16. What is the sum of all possible lengths of the hypotenuse of ABC ?

Proposed by: Kaining

Answer: $\boxed{43}$

Solution: The Pythagorean triples with all sides under 16 are $(3, 4, 5)$ and $(5, 12, 13)$. $(6, 8, 10)$ and $(9, 12, 15)$ also work, so we have $5 + 13 + 10 + 15 = \boxed{43}$.

Problem 4. Let square $ABCD$ be inscribed in circle ω_1 , and let circle ω_2 be inscribed in $ABCD$. Given that $AB = 2$, what is the difference between the areas of ω_1 and ω_2 ?

Proposed by: Kaining Zhang

Answer: $\boxed{\pi}$

Solution: Let O be the center of $ABCD$. Since $ABCD$ is inscribed in w_1 , the radius of w_1 must be $OA = \frac{AC}{2} = \frac{2\sqrt{2}}{2} = \sqrt{2}$, so its area is $(\sqrt{2})^2\pi = 2\pi$. Since w_2 is inscribed in $ABCD$, its radius is half the side length of $ABCD$, so its radius is 1, and its area is $(1)^2\pi = \pi$. Thus, our difference is $2\pi - \pi = \boxed{\pi}$.

Problem 5. $10!$ can be written as $2^a 3^b 5^c 7^d$. What is $a + b + c + d$?

Proposed by: Kaining Zhang

Answer: $\boxed{15}$

Solution: Legendre's formula states that the exponent of prime p in the prime factorization of $n!$ is $\lfloor \frac{n}{p} \rfloor + \lfloor \frac{n}{p^2} \rfloor + \lfloor \frac{n}{p^3} \rfloor + \dots$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x .

For $p = 2$: $\lfloor \frac{10}{2} \rfloor + \lfloor \frac{10}{4} \rfloor + \lfloor \frac{10}{8} \rfloor = 5 + 2 + 1 = 8 = a$.

For $p = 3$: $\lfloor \frac{10}{3} \rfloor + \lfloor \frac{10}{9} \rfloor = 3 + 1 = 4 = b$.

For $p = 5$: $\lfloor \frac{10}{5} \rfloor = 2 = c$.

For $p = 7$: $\lfloor \frac{10}{7} \rfloor = 1 = d$.

Therefore we have $8 + 4 + 2 + 1 = \boxed{15}$.

Problem 6. Let p and q be the roots of the quadratic $2x^2 - 7x + 3$. What is $p^2 + q^2$? Express your answer as a common fraction in simplest form.

Proposed by: Kaining Zhang

Answer: $\boxed{\frac{37}{4}}$

Solution: By Vieta's, $p + q = \frac{7}{2}$, and $pq = \frac{3}{2}$. Thus,

$$(p + q)^2 - 2pq = \left(\frac{7}{2}\right)^2 - 2\left(\frac{3}{2}\right) = \frac{49}{4} - 3 = \boxed{\frac{37}{4}}.$$

Problem 7. Earl and Frank are playing a game. They each roll a fair 6-sided die. Earl wins if the product of their rolls has an odd number of divisors; Frank wins otherwise. If they play 36 independent games, what is the expected number of games Earl will win?

Proposed by: Alice Lee

Answer: $\boxed{8}$

Solution: We would like to calculate the probability that Earl will win any one game. Notice that Earl wins if the product of the 2 numbers has an odd number of factors, meaning he wins if the product of the 2 numbers is a perfect square. We must calculate the probability of $m \cdot n$ being a perfect square, given that $m \in \{1, 2, 3, 4, 5, 6\}$ and $n \in \{1, 2, 3, 4, 5, 6\}$, and we can do so using casework:

Case 1: $m \cdot n = 1^2$ only occurs when $(m, n) = (1, 1)$.

Case 2: $m \cdot n = 2^2$ occurs when $(m, n) = (2, 2)$, $(m, n) = (1, 4)$, and $(m, n) = (4, 1)$.

Case 3: $m \cdot n = 3^2$ only occurs when $(m, n) = (3, 3)$.

Case 4: $m \cdot n = 4^2$ only occurs when $(m, n) = (4, 4)$.

Case 5: $m \cdot n = 5^2$ only occurs when $(m, n) = (5, 5)$.

Case 6: $m \cdot n = 6^2$ only occurs when $(m, n) = (6, 6)$.

There are 8 pairs for $m \in \{1, 2, 3, 4, 5, 6\}$ and $n \in \{1, 2, 3, 4, 5, 6\}$ such that $m \cdot n$ is a perfect square. There are 36 possible pairs for (m, n) . Thus, Earl's probability of winning an individual game is $\frac{8}{36}$.

To calculate Earl's expected win number of wins after 36 games, we can simply multiply $\frac{8}{36} \cdot 36 = 8$, as each game is independent of the next. Thus, our answer is $\boxed{8}$.

Problem 8. Angie, Alice, and Sneha each pick 3 distinct positive integers less than 12. Sneha knows Angie's number, but nobody else knows the other numbers that are chosen. Sneha claims that her number to the power of Angie's number ends in 3. Upon hearing this, Alice claims that her number to the power of Angie's number ends in 7. What is the sum of all possible products of the 3 numbers?

Proposed by: Sneha Kundu

Answer: $\boxed{546}$

Solution: We start by considering the possibilities of the last digit of Sneha and Alice's numbers. Cycling through all the last digits from 0-9, it becomes clear that Sneha and Alice's numbers must be 3 and 7 in some order. This automatically satisfies the condition in the problem since $3^n \equiv 3 \pmod{10}$ and $7^n \equiv 7 \pmod{10}$ when $n \equiv 1 \pmod{4}$ and $3^n \equiv 7 \pmod{10}$ and $7^n \equiv 3 \pmod{10}$ when $n \equiv 3 \pmod{4}$. Now, we divide into cases.

Case 1: Sneha's number is 3 and Alice's number is 7. Then, Angie's number must be 1 mod 4 so Angie's number could be 1, 5, or 9.

Case 2: Sneha's number is 7 and Alice's number is 3. Then, Angie's number must be $3 \pmod{4}$ so Angie's number could only be 11.

This means the sum of all possible products of the 3 numbers is $(3 \cdot 7)(1 + 5 + 9 + 11) = 3 \cdot 7 \cdot 26 = \boxed{546}$.

Problem 9. Cyclic quadrilateral $ABCD$ is inscribed in $\odot O$ with AC as a diameter. $\overline{BC}$ and $\overline{AD}$ are extended past B and A , respectively, intersecting outside the circle at E . Given that $AB = AD = 21$ and $BE = 72$, find the length of BD . Express your answer as a common fraction in simplest form.

Proposed by: Alice Lee

Answer: $\boxed{\frac{168}{5}}$

Solution: Because $\overline{AC}$ is the diameter of $\odot O$, $\angle ADC = \angle ABC = \angle EBA = 90^\circ$. Observe $\triangle EBA$ is a right triangle with leg lengths of $EB = 72$ and $AB = 21$. This is similar to a $7 - 24 - 25$ right triangle, so $EA = 3 \cdot 25 = 75$.

By the Power of a Point theorem, $EA \cdot ED = EB \cdot EC$. Therefore, $75 \cdot (75 + 21) = 72 \cdot EC$, and $EC = \frac{75 \cdot 96}{72} = 100$. Because $EC = 100$ and $EB = 72$, notice $BC = 100 - 72 = 28$.

Next, because $AC = AC$ and $AB = AD$, by HL congruence, $\triangle ABC \cong \triangle ADC$. As a result, $BC = DC = 28$. Also, $\triangle ABC$ is similar to a $3 - 4 - 5$ right triangle, so $AC = 5 \cdot 7 = 35$.

Finally, we can find the length of BD using Ptolemy's Theorem. We know $BD \cdot AC = BC \cdot AD + AB \cdot CD$, so $BD \cdot 35 = 28 \cdot 21 + 21 \cdot 28$. Therefore, $BD = \frac{28 \cdot 21 \cdot 2}{35} = \frac{8 \cdot 21}{5} = \boxed{\frac{168}{5}}$.

Problem 10. A group of eight distinct friends bought a cheese pizza and pepperoni pizza, each with eight slices. The slices are divvied up among them so that each friend gets two slices. Assuming slices with the same topping are indistinguishable, in how many ways can this happen?

Proposed by: Samuel Kretzschmar

Answer: $\boxed{1107}$

Solution: We can use casework on the number of people getting two pepperoni slices. For every person who gets two pepperoni slices, another must get two cheese slices (zero pepperoni) to keep the total number of pepperoni slices given at 8. The rest must get one slice of each. We can then choose the individuals with zero, one, and two pepperoni slices for each case.

No two slices: All 8 people get one slice of each, only 1 way.

One person with two pepperoni slices: 1 two slices, 1 zero slices, 6 one slice for a total of $\frac{8!}{6!1!1!} = 56$.

Two people with two slices: 2 two slices, 2 zero slices, 4 one slice for a total of $\frac{8!}{2!2!4!} = 420$.

Three people with two slices: 3 two slices, 3 zero slices, 2 one slice for a total of $\frac{8!}{3!3!2!} = 560$.

Four people with two slices: 4 two slices, 4 zero slices, 0 one slice for a total of $\frac{8!}{4!4!0!} = 70$.

The total is thus $1 + 56 + 420 + 560 + 70 = \boxed{1107}$.
